Bluestock Mutual Fund Analytics

Date: June 24, 2026

1. Project Scope & Architecture

The mutual fund analytics engine was designed using a robust ETL pipeline. Data was ingested from raw CSVs, rigorously cleaned (forward-filling NAV holidays, validating numeric constraints), and loaded into a 6-table SQLite Star Schema. The dimensional model separates fact tables (NAV, transactions, AUM, performance) from dimensions (fund, date) to optimize query performance.

2. Quantitative Metrics & Optimization

Advanced financial models were deployed to assess fund viability. We computed annualized risk-adjusted returns using the Sharpe and Sortino ratios, derived Alpha/Beta via OLS regression against the Nifty 50, and measured downside risk using 95% Historical VaR.

Furthermore, we executed 1,000-scenario Monte Carlo simulations using Geometric Brownian Motion to project 5-year NAV growth, and solved for the Markowitz Efficient Frontier to find the optimal maximum Sharpe ratio weights for our top 5 funds.

3. Deliverables Status

- D1 ETL Pipeline: Completed (etl_pipeline.py)
- D2 SQLite Database: Completed (bluestock_mf.db, strictly .gitignored)
- D3 EDA Notebook: Completed
- D4 Performance Metrics: Completed
- D5 Interactive Dashboard: Completed (Streamlit Web App)
- D6 Advanced Analytics: Completed
- D7 Presentations: Completed (This Report & PPTX)
- Bonus 1: Cron Job Batch Script (schedule_etl.bat)
- Bonus 2: Streamlit Dashboard Overhaul
- Bonus 3: Monte Carlo Simulation (Notebook generated)
- Bonus 4: Markowitz Portfolio Optimization (Notebook generated)
- Bonus 5: Automated HTML Email Report Script